# Ryan Caudill

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## EDUCATION

**University of South Carolina** *Columbia, SC*
**B.S. in Computer Science |** Artificial Intelligence Concentration | Spanish Minor **Graduation May 2026**
**Awards:** Dean's List, Capstone Scholar Major GPA: 3.55

## WORK EXPERIENCE

**Oracle Cloud Infrastructure** *Nashville, TN*
***Software Engineer*** **Jul 2026 – Present**

- Incoming July 2026

**University of South Carolina** *Columbia, SC*
***Undergraduate Research Assistant*** **Oct 2025 – Jun 2026**

- Optimized multi-view data capture by implementing ROS2, hardware triggers, and multiprocessing on an Nvidia Jetson Orin, successfully synchronizing four camera streams at the hardware maximum of 20 Hz.
- Designed and deployed a Qt-based GUI in Ubuntu, streamlining the researchers' workflow by enabling rapid switching between software builds and significantly accelerating the data collection process.

***Undergraduate Teaching Assistant*** **Jan 2024 – Dec 2025**

- Designed and implemented comprehensive system architectures for a Language Learning Application, Degree Planning Software, Music Creation Program, and Digital Escape Room in JavaFX
- Provided strategic leadership to 32 development teams, setting a clear direction and fostering collaboration to meet key project milestones effectively and on time.
- Facilitated detailed bi-weekly code reviews as part of 2-week sprint cycles, ensuring high-quality code, adherence to best practices, and consistent progress toward project objectives.

**South Carolina Department of Transportation** *Columbia, SC*
***Software Engineer Intern*** **May 2025 – Aug 2025**

- Automated OSHA injury report submissions by transforming SQL Server data into JSON and integrating with the OSHA ITA API, cutting report time from 7 days to 15 seconds
- Developed a Blazor web app with spreadsheet-style SQL Server editing, improving data accuracy and enabling faster updates by non-technical staff.
- Built a LangChain demo in Python showcasing LLM Agents for natural language SQL query generation.
- Maintained and modernized legacy web applications, improving usability for internal teams.

## PROJECTS

***Dermalite***

- Trained three distinct skin lesion classification models using PyTorch, engineering a custom ResNet50 CNN by replacing the final fully connected layer with a Random Forest classifier.
- Converted and quantized the models using CoreML to deploy the classifiers for real-time, on-device processing.
- Achieved 90% sensitivity and 60% specificity, strategically optimizing the model to minimize false negatives for safer clinical diagnostic screening.

***GoSynchro.io***

- Engineered the core back-end infrastructure using Node.js and TypeScript, developing scalable, high-performance APIs to support the cross-platform React Native mobile application.
- Directed project DevOps by designing and implementing CI/CD pipelines utilizing GitHub Actions and Azure, streamlining deployment workflows, and ensuring reliable builds.

***TA Auto grader***

- Built an autograding platform with a Django frontend, FastAPI backend, PostgreSQL, and a CI/CD pipeline to analyze student GitHub repositories for software engineering assignments.

## SKILLS

***Tools:*** Git/GitHub, MATLAB, Docker, PyTorch, Azure, Raspberry Pi, Ubuntu, Hadoop, GitHub Actions
***Languages:*** Java, Python, C++, C#, Haskell, Bash, Swift, TypeScript, Pig
***Frameworks:*** FastAPI, JavaFX, Django, .NET, React Native, Expo, ROS, Qt